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Dual-domain solvent-locked electrolyte enabled durable

4.5 V-class sodium batteries

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Abstract

Sodium-ion batteries hold significant promise for sustainable energy storage by addressing resource scarcity and safety concerns. Elevating the upper-cut-off voltage is crucial for sodium-ion batteries to maximize electrode capacity and narrow their energy-density gap with commercial lithium-ion batteries. However, conventional electrolytes exhibit electrochemical instability on the highly desodiated positive electrode surface, undergoing strongly electrophilic attack and continuous decomposition that results in oligomer-rich interphases and rapid capacity fading. Here, we design a solvent-locked carbonate electrolyte creating electrochemically stable solvent-reinforced solvation structure and anion-rich interfacial shield-derived boride-/fluoride-rich interphase on positive electrode, thereby suppressing current leakage and parasitic reactions. The tailored electrolyte exhibits good compatibility with commercialized oxides and polyanionic positive electrodes. As-assembled Na||Na_{2.26}Fe_{1.87}(SO₄)₃ cells deliver extended lifespans operating to 4.5 V, retaining 88.2% capacity after 16,500 cycles at 1000 mA g⁻¹ (coin cell) and 93.9% after 500 cycles at 100 mA g⁻¹ (pouch cell). This work will inspire durable electrolyte and interphase design for high-energy batteries and beyond.

Introduction

Drastic artificial intelligence (AI) development and the extensive exploitation of renewable energy sources have intensified the need for grid-scale energy storage systems. Abundant sodium-ion batteries (SIBs) have garnered significant attention and achieved substantial industrial progress, overcoming sustainable supply constraints from lithium scarcity and price volatility.¹ Advanced SIBs integrating metallic sodium negative electrode and high-capacity positive electrode are anticipated to deliver an energy density comparable to that of commercial lithium-ion batteries with long cycling durability.² However, complete Na^+ extraction from positive electrode crystals necessitates high operation voltage, imposing severe electrolyte compatibility challenges.³ During deep desodiation, electron removal from the e_g -orbital of redox couple induces severe surface hole accumulation and drastic Fermi-level downshifting, creating a strongly electrophilic positive electrode surface that initiates electrophilic attacks on free solvents and Na^+ -solvation complexes.⁴ Solvent-dominated decomposition generates organic radicals forming oligomer-rich interphases through electrophilic addition-condensation reactions. These porous interphases fail to suppress electron tunneling and block solvent permeation, thereby accelerating continuous electrolyte depletion and capacity fading.⁵ Moreover, conventional carbonate exhibit limited anodic stability under high voltage, releasing gases (e.g., O_2 , CO_2) that propagate parasitic reactions and safety hazards.⁶

Electrolyte engineering promises to address high-voltage positive electrode/electrolyte interfacial instability. Current approaches focus on unilateral regulation. For instance, bulk solvation control via high-concentration or locally concentrated electrolytes to form solvent-separated contact ion pairs or ion aggregates; interfacial design using fluorine-containing additives (e.g., fluoroethylene carbonate, fluorobenzene) for sacrificial preformed interphases.⁷ While analogous to localized biological protection, these approaches incur significant trade-offs under practical applications including compromised ionic conductivity, increased viscosity, and negative electrode incompatibility.⁸ Fundamentally, interfacial oxidation follows an adsorption-mediated preferential decomposition mechanism where solvents experience competing electrostatic forces, the intense Helmholtz plane potential versus Born solvation energy constraints.⁹ This dynamic bulk-interface coupling necessitates synergistic regulation on bulk solvation structures to minimize high-energy conformations and directed interfacial assembly through anion-prioritized decomposition kinetics.¹⁰ Furthermore, elucidating the passivation behavior of interphase components, particularly their voltage-dependent electronic/ionic transport properties, is equally important for comprehensive battery stability.

In this work, a solvent-locked electrolyte is designed to address its electrochemical instability at high voltage. The symmetric difluoro(oxalato)borate anion (DFOB^-), employed as an additive to the industrial electrolyte (1M NaPF_6 in ethylene carbonate (EC)/ethyl methyl carbonate (EMC)), exhibits weak coordination with carbonate solvents but strong binding affinity towards Na^+ -exposed positive electrode surface. Leveraging steric hindrance and electron-withdrawing effects of DFOB^- anion, the tailored electrolyte achieves effective locking to solvent molecules at dual domains of bulk-phase electrolyte (domain I) and positive electrode interface (domain II). We therefore designate it as a dual-domain solvent-locked electrolyte (DDSLE). Crucially, the preferential decomposition of DFOB^- generates boride/fluoride-rich CEI components with wide bandgaps (>6.2 eV), providing electron insulation and improved oxidation resistance. Consequently, DDSLE electrolyte exhibits good cycling stability in 4.5 V-class $\text{Na}||\text{Na}_{2.26}\text{Fe}_{1.87}(\text{SO}_4)_3$ coin and pouch cells.

Results and Discussion

The DDSLE electrolyte with high-voltage stability is engineered from an industrial carbonate baseline (BE: 1M

NaPF₆ EC/EMC). While BE electrolyte offers favorable ionic conductivity, low viscosity, and separator wettability, it exhibits poor high-voltage stability (~ 3.4 V vs. Na⁺/Na).¹¹ Introducing low-concentration NaDFOB slightly reduced conductivity due to its lower intrinsic conductivity versus NaPF₆, hindered ion migration, and increased viscosity (Fig. 1a and Supplementary Figs. 1–2). Importantly, at this concentration (0.2M), NaDFOB addition fails to satisfactorily enhance the electrolyte's anodic stability limit, primarily because it cannot sufficiently control the structure within the bulk electrolyte and at the positive electrode interface (Supplementary note 1). Higher NaDFOB concentrations modestly recovered conductivity and further increased viscosity, but beyond 0.6M, electrolyte cloudiness indicated salt precipitation. Thus, 0.4M NaDFOB was selected as optimal for the DDSLE electrolyte (1M NaPF₆ + 0.4M NaDFOB in EC/EMC). The solvation restructuring during DDSLE preparation was examined using *in situ* Fourier transform infrared spectroscopy (FTIR). Its setup enables controlled injection of salts and solvents, with a standing time of 1 hour to ensure full salt dissolution and solvation equilibrium. The free C=O stretches were observed at 1800 cm⁻¹ for EC and 1740 cm⁻¹ for EMC, while the free C–O vibration of EMC was found at 1260 cm⁻¹. In contrast, the coordinated C=O vibrations for EC and EMC appeared at 1770 cm⁻¹ and 1720 cm⁻¹, respectively, with a coordinated C–O band for EMC at 1300 cm⁻¹ (Figs. 1b–c). In the DDSLE electrolyte, blueshifts were observed for free species: the EC C=O peak shifted from 1800 to 1805 cm⁻¹ and the EMC C–O–C peak from 1270 to 1274 cm⁻¹, indicating a modified electronic environment and stronger intermolecular interactions. A blueshift was also evident for coordinated EMC (C–O–C from 1300 to 1303 cm⁻¹), confirming enhanced EMC–Na⁺ coordination (Supplementary Fig. 3). These shifts result from electron donation from solvent oxygen lone pairs to Na⁺ vacant orbitals, which reduces electron density at oxygen, increases bond force constants, and raises vibrational frequencies.¹²

Solvent-reinforced solvation structure in DDSLE electrolyte and its improved antioxidant stability is quantitatively analyzed. Although the overall Na⁺ solvation shell composition remained similar after NaDFOB addition (Figs. 1d–e and Supplementary Data 1,2), quantitative analysis revealed distinct solvent-reinforced structure: the coordination number of PF₆⁻ decreased from 1.46 (BE electrolyte) to 1.28 (DDSLE electrolyte), with EC increasing slightly from 1.49 to 1.63 and EMC constant at 2.77 (Fig. 1f and Supplementary Fig. 4). DFOB⁻ anion shows strongly non-solvating effect, evidenced by the large Na–F (DFOB⁻) distance (7.45 Å), which compresses the primary solvation shell.¹³ Consequently, key bond lengths between Na⁺ and neighboring molecules shortened, including Na–O (EC) from 2.35 Å to 2.33 Å, Na–O (EMC) from 2.43 Å to 2.36 Å, and Na–F (PF₆⁻) from 2.58 Å to 2.48 Å (Fig. 1g). Density functional theory (DFT) calculation linked this structural tightening to electronic stabilization, showing that the reduced average coordination length lowered the highest occupied molecular orbital (HOMO) energy from –7.92 eV to –8.22 eV by disrupting π -electron delocalization in the carbonate solvents (Fig. 1h, Supplementary Fig. 5 and Supplementary Data 3).¹⁴ Moreover, the compacted solvation cluster decreased the hydrodynamic radius, increasing the Na⁺ transference number from 0.391 ± 0.004 to 0.474 ± 0.048 (Fig. 1i and Supplementary Fig. 6). NMR spectra further confirmed the solvent-reinforced solvation structure in DDSLE electrolyte, with upfield shifts observed for ²³Na (–0.44 ppm) and ¹⁷O in EC carbonyl (–1.25 ppm), indicating increased electron density around Na⁺ and stronger solvent coordination (Fig. 1j). Concurrently, minor upfield shifts (–0.02 and –0.03 ppm) for ¹⁹F indicates increased electron density around PF₆⁻ anion, consistent its reduced coordination in the solvation shell. Raman spectroscopy quantified a shift from contact ion pairs (CIP, 33.3%) in BE electrolyte to a higher proportion of solvent-separated ion pairs (SSIP, 45.4%) in the DDSLE electrolyte (Fig. 1k). These changes originate from the strong electron-withdrawing and steric nature of DFOB⁻ anion, which intensifies the local electric field, drives electron density redistribution, and stabilizes a solvent-reinforced solvation structure, thereby improving the electrolyte's overall electrochemical stability and transport performance.^{15,16} In terms of

fundamental design and mechanism, the "solvent-reinforced" and "weakly solvating" strategies are not opposing but represent two distinct, effective pathways in electrolyte engineering (Supplementary note 2). The former is analogous to "consolidating internal affairs," enhancing intrinsic stability by strengthening internal bonds. The latter resembles "skillful diplomacy," guiding the formation of an ideal interface by optimizing reaction sequence and kinetics.

The positive electrode surface, being the primary site for electrolyte adsorption and reaction, necessitates careful electrolyte design aligned with its charge characteristics to enable effective interfacial passivation. The alluaudite-type $\text{Na}_{2.26}\text{Fe}_{1.87}(\text{SO}_4)_3$ (NFS) positive electrode delivers a high average discharge voltage (3.8 V) and reversible specific capacity within 2.0–4.5 V.¹⁷ Detailed structural characteristics are provided in Supplementary Fig. 7 and Supplementary table 1. The (200) plane with well-ordered monoclinic atomic arrangement, identified as the dominant exposed facet via XRD refinement and high-resolution transmission electron microscopy (HRTEM) (spacing: 5.54 Å; C2/c space group, Fig. 2a), was selected as substrate model for simulating electrolyte contact due to its structural sensitivity and signal clarity.¹⁸ Electron density mapping showed localized high-density regions at coordinated oxygen sites, while the overall surface exhibits negative charge character due to high Na content (Fig. 2b and Supplementary Data 4). AIMD simulations of the Na-exposed NFS (200) model demonstrated strong DFOB^- adsorption (−6.778 kJ/mol) compared to PF_6^- (−5.547 kJ/mol), EC (−1.067 kJ/mol), and EMC (−0.877 kJ/mol), forming a dense anion-rich interfacial shield that isolates the positive electrode from bulk electrolyte (Figs. 2c–d, Supplementary Fig. 8 and Supplementary Data 5,6).¹⁹ Component distribution profile along the *c*-axis confirmed DFOB^- enrichment near the positive electrode/electrolyte interface. Furthermore, electrochemical quartz crystal microbalance (EQCM) measurements revealed a rapid decrease in the negative-mass-resonance frequency of NFS particles upon 1309-second contact with the DDSLE electrolyte (−151.1 Hz for BE; −338.8 Hz for DDSLE; Fig. 2e), directly evidencing accelerated DFOB^- adsorption (rapid mass increase) and anion-rich interfacial shield formation.²⁰

Preferential oxidation reaction of anion-rich interfacial shield is demonstrated by DFT calculation and *in situ* electrochemical characterizations. DFOB^- possesses the highest HOMO level (−7.61 eV) and the lowest oxidation stability compared to other components (e.g., PF_6^- at −10.23 eV; EC at −8.40 eV; EMC at −8.08 eV) (Fig. 2d),²¹ enabling its preferential oxidation on positive electrode surface. *In situ* Raman spectroscopy reveals emerging peaks at 469–1240 cm^{-1} during initial charging (3.3–3.7 V), corresponding to B/P/C–O/F species formation from DFOB^- oxidative decomposition (Fig. 2f). Upon charging to 4.2 V, significant peak changes at 2780–2960 cm^{-1} indicate H–O-rich species from solvent decomposition. In contrast, cells with BE electrolyte exhibited distinct peaks at 1316–1627 cm^{-1} beyond 3.9 V and strong peaks at 2255–2779 cm^{-1} above 4.2 V, confirming severe solvent decomposition occurring at significantly lower voltages.²² *In situ* differential electrochemical mass spectrometry (*In situ* DEMS) further corroborates the analysis from the perspective of gas evolution. For DDSLE-involved cells, CO_2 evolution initiates at 2.9 V (charging), intensifies by 3.9 V, dips then rise again above 4.3 V, with minimal CO/CH_4 (Fig. 2g). This confirms DFOB^- decomposition (2.9–4.3 V) as the primary CO_2 source before solvent decomposition (>4.3 V).²³ BE-involved cells display earlier CO_2 onset (2.7 V) and recurring CO_2/CO peaks at 3.7 V (discharge) due to unshielded interface structures. The more pronounced overcharge phenomenon can be attributed to inherent structural instabilities combined with harsh operating conditions of *in situ* DEMS cell, which collectively result in degraded interfacial properties and poor kinetic behaviors (Supplementary Fig. 9 and note 3). Critically, multi-cycle DEMS quantification (Fig. 2h) reveals DDSLE generates significant gases only in initial cycle (H_2 : 1.41, CO_2 : 0.96, CO : 0.05, CH_4 : 0.07 μmol), which becomes negligible in subsequent cycles.²⁴ Conversely, BE-based cells exhibit substantial initial-cycle gases (H_2 : 2.48, CO_2 : 0.74, CO : 0.43 μmol) with increasing amounts in later cycles. These results collectively demonstrate that the anion-rich shield-derived passivation layer effectively suppresses persistent electrolyte

decomposition at high voltage.²⁵ The preferential oxidation of DFOB⁻, indicated by minor current peaks at 3.0–3.4 V vs. Na⁺/Na, elevates the anodic stability limit of electrolytes from 3.4 V vs. Na⁺/Na (BE electrolyte) to 5.1 V vs. Na⁺/Na (DDSLE electrolyte, Fig. 3a). This enhanced oxidative stability is further confirmed through statistical data and tests on additional substrate (Ni foil) in Na||inert metal coin cells (Supplementary Fig. 10).

Oxidational decomposition pathways of the electrolyte are clarified via combined *in situ* Raman spectroscopy and gas evolution (Figs. 3b–c), providing mechanistic insights based on electron transfer and molecular orbital interactions. Under anodic polarization, DFOB⁻ anions undergo preferential adsorption via electrostatic forces and potential orbital coupling with the positive electrode surface. Oxidation initiates at 2.8–3.4 V through electron transfer from the HOMO (localized on oxalate oxygen lone pairs and B–F bonds) of DFOB⁻ to positive electrode unoccupied states, generating an electron-deficient species that undergoes nucleophile-facilitated or spontaneous C–C cleavage of its oxalate moiety, releasing CO₂ and forming electrophilic BF₂CO₂⁻ intermediates.²⁶ These boron-centered electrophiles participate in competitive pathways: (1) nucleophilic attack by Na⁺ (Lewis acid coordination) triggers O=C=O scission, releasing CO and forming NaBOF₂, which hydrolyzes to NaF and B₂O₃; (2) O substitution by F⁻ (from PF₆⁻ decomposition) displaces CO₂, yielding BF₃ that further reacts with NaF to form stable NaBF₄. At higher potentials (>3.9 V vs. Na⁺/Na), potential-intensified electron tunneling oxidizes ester solvents: EC undergoes HOMO electron removal (carbonyl oxygen), generating a radical cation (EC^{•+}) that suffers nucleophilic ring-opening at its electrophilic carbonyl carbon, followed by β -scission of the linear radical to release CO₂, CO, and CH₄ via radical recombination/H-abstraction.²⁷ In the case of EMC, the cleavage of the C–O bond is facilitated by electron transfer into the low-lying σ^* orbital, a process enhanced under high voltage. Decomposition proceeds primarily through C–O bond scission, releasing CO₂ and various organic fragments.²⁸

The electronic origin of anion-derived insoluble enhancing the antioxidation stability at positive electrode/electrolyte interface was rigorously evaluated through integrated theoretical and experimental analysis, with electron tunneling resistance identified as the primary determinant for electrochemical stability at high potential (Supplementary Fig. 11 and note 4). Among reported insoluble compounds, borides and fluorides exhibit wide bandgap values (>6.2 eV), correlating directly with enhanced dielectric strength against electron tunneling at elevated potentials (Fig. 3d, Supplementary Fig. 12 and Supplementary table 2).²⁹ This structure-property relationship was quantitatively established through linear sweep voltammetry on material-specific working electrodes, demonstrating a definitive positive correlation ($R^2 > 0.95$) between experimental decomposition voltage and theoretical bandgap. The empirical relationship confirms that inorganic phases, particularly borides and fluorides, are essential for maintaining electrochemical integrity above 4.0 V, whereas organic polyesters, despite moderate intrinsic insulation capability, suffer from inherent limitations, including low ionic conductivity, structural porosity enabling electrolyte penetration, and progressive dissolution during cycling that compromises interfacial cohesion. Furthermore, bond valence site energy (BVSE) calculations indicate that the Na⁺ conductivity of these boride- and fluoride-based phases is comparable to that of typical inorganics like Na₂O, supporting their role in facilitating ion transport while blocking electron leakage (Supplementary Fig. 13).³⁰ This is corroborated by leakage current measurements on the positive electrode surface. At voltages exceeding 4.5 V, cells employing DDSLE electrolyte exhibit a leakage current reduction of 34.6%–79.2% compared to the BE (Fig. 3e and Supplementary Fig. 14), demonstrating that the boride/fluoride-rich interphase effectively suppresses electron tunneling and enhances interfacial stability.³¹

Boride/fluorine-rich CEI induced by anion-rich interfacial shield is confirmed through XPS spectra. The cycled NFS positive electrode obtained from DDSLE shows a strong fluorine signal (attributed to P–F and NaF species) and abundant boron-based species (B–F, B–O, Na–B–O).³² In contrast, the positive electrode from the BE electrolyte

shows negligible boron content but significant organic species, indicative of continuous electrolyte decomposition due to an ineffective CEI barrier (Fig. 4a, Supplementary Figs. 15–16).³³ TOF-SIMS depth profiling delineated a stratified structure within the DDSLE-based CEI: boron concentration peaked near the surface (~ 10 s sputtering time) before declining, while F^- and Na^+ maintained uniform distributions throughout the film (Fig. 4b). Organic fragments (e.g., $C_2H_2O^-$) were predominant only at the outermost surface (< 5 s sputtering) and decreased sharply with depth. These findings confirm the formation of an inorganic-rich (boride/fluoride-dominated), compositionally graded CEI structure in DDSLE-based cells. During the continuous electrochemical cycling process, these borides and fluorides were well preserved and continuously acted as a passivation layer at the interface (Supplementary Fig. 17). Consequently, the DDSLE-generated CEI possessed a homogeneous, low surface potential centered at ~ 1.2 V, demonstrating efficient charge dissipation capability (Figs. 4c–d).³⁴ Conversely, the BE-derived interface exhibited a heterogeneous potential field (0.4–1.4 V range), indicating localized charge accumulation hotspots that promote electronic tunneling and persistent interfacial side reactions.

Cryogenic transmission electron microscopy (cryo-TEM) images of the cycled NFS positive electrode showed clear lattice fringes corresponding to the (200) crystal plane, indicating that its overall structure was well preserved (Figs. 4e–f). BE electrolyte formed a thick and uneven CEI (average thickness: 28.3 ± 3.4 nm), whereas the DDSLE formed a significantly thinner and uniform CEI (average thickness: 13.6 ± 0.9 nm).³⁵ These inhomogeneities were also evident in SEM images, which revealed rough deposits on the electrode surface consistent with the formation of electrolyte decomposition by-products (Supplementary Fig. 18). Atomic force microscopy (AFM) directly quantified the resulting surface roughness (Fig. 4g). BE-involved particles exhibited high inhomogeneity within a $1 \mu m^2$ area (maximum depth variation: $0.89 \mu m$), while the DDSLE-derived interface film was smoother and more uniform (maximum depth variation: $0.64 \mu m$).³⁶ The thinner CEI achieved with DDSLE facilitates shorter-distances and rapid ion transport across interface.

Mechanical stability in DDSLE-derived CEI during 100 cycles was elucidated through EQCM (Fig. 4h). Both systems showed an initial linear mass increase ($\sim 0.69 \mu g cm^{-2}$ or DDSLE, $\sim 0.93 \mu g cm^{-2}$ for BE over initial 30 cycles). Critically, from cycle 30 onwards, the DDSLE-derived CEI stabilized, maintaining minimal mass changes (0.72 – $0.73 \mu g cm^{-2}$ per subsequent 30-cycle interval). In stark contrast, the BE-derived CEI mass increased rapidly (e.g., $1.71 \mu g cm^{-2}$ between cycles 29–31, $1.07 \mu g cm^{-2}$ between cycles 79–81), indicating CEI rupture and ongoing electrolyte decomposition (Fig. 4h).³⁷ Concurrently, the energy dissipation per cycle (related to CEI viscoelasticity) increased by only 1.1 units for DDSLE-involved cell over 100 cycles, compared to 1.6 units for BE-involved cell. This lower dissipation increase signifies a more rigid interface. Analysis of mass change per unit dissipation energy revealed that the initially low mass-to-dissipation ratios for both electrolytes indicate predominant anion-driven CEI formation during early cycling stages (Fig. 4i). Distinct near-vertical profile in early cycles of DDSLE-involved cell is attributed to the preferential decomposition of $DFOB^-$ anion, forming a boride/fluorine-rich CEI (inset of Fig. 4i). By cycle 11, DDSLE-involved cell maintained a low mass/dissipation ratio, consistent with $DFOB^-$ adsorption continuously repairing defects in the existing inorganic matrix. In contrast, BE-involved cell exhibited a high mass-to-dissipation ratio, indicative of sustained solvent decomposition that yields soft, porous organic species. This structurally deficient layer fails to passivate the electrode surface, permitting continuous electron tunneling across the interface and thereby driving persistent electrolyte consumption and progressive CEI thickening.³⁸ While the boride/fluorine-rich CEI with dense structure effectively suppresses electron tunneling, enables homogeneous charge distribution, and provides mechanical strength, significantly enhancing cell stability.

The stable interphase chemistry on Na negative electrode in DDSLE electrolyte was investigated. $DFOB^-$ anion

exhibits strong adsorption energy on Na metal surface ($-5.98 \text{ kJ mol}^{-1}$ vs. $-0.37 \sim -0.73 \text{ kJ mol}^{-1}$ for solvents). Coupled with its low-lying LUMO energy level, DFOB⁻ easily accepts electrons from Na negative electrode, leading to preferential reductive decomposition (Fig. 5a and Supplementary Data 7). Optical microscopy of symmetric Na cells reveals dense microbubbles within initial 20 minutes of Na plating/stripping, corresponding to DFOB⁻ reduction (Fig. 5b). These bubbles dissipate without sustained growth, confirming rapid interface stabilization.³⁹ Conversely, BE electrolyte shows persistent bubble evolution on bare Na due to absent protective interphase. These results demonstrate DFOB⁻-derived interfacial shield formation on negative electrodes is essential for achieving durable passivation.

The components and structure of formed SEI in DDSLE electrolyte were further investigated. After initial five cycles, the SEI formed in BE electrolyte exhibits a broad thickness distribution ranging from 16.4 nm to 57 nm, whereas the SEI in DDSLE electrolyte is more uniform, with thickness confined between 24.1 nm and 27.8 nm (Fig. 5c). This marked heterogeneity in the BE-derived SEI is attributed to vigorous side reactions at the Na negative electrode, as visualized by in situ optical microscopy. XPS analysis (B 1s, O 1s, F 1s spectra) of the SEI directly demonstrates that the DDSLE-derived SEI is rich in boron-containing species (B–O, Na–B–O) and fluorides (B–F, NaF), while the BE-based SEI shows no detectable boron and only minor NaF content (Fig. 5d). Depth profiling revealed that in the DDSLE system, the B content is relatively low at the outermost surface ($\sim 0.68\%$) but increases sharply after etching and stabilizes at $\sim 2.3\%$ (Fig. 5e).⁴⁰ Similar trends are observed for P, F, and Na, with contents stabilizing around 1%, 9%, and 33%, respectively. In the BE system, both P and F remain low ($<0.4\%$ and $<1\%$), while Na exhibits an increasing gradient from surface to bulk, consistent with a non-uniform SEI structure. C element shows a decreasing trend from surface to interior in both systems, indicating outer-rich organic distribution. Notably, O element in DDSLE remains largely uniform with depth, while in BE it increases significantly toward the interior. These findings establish DDSLE-derived boride/fluorine-rich, homogeneous SEI versus BE-derived oligomer-rich, heterogeneous structure.⁴¹ Correspondingly, a lower corrosion current density for the DDSLE system (0.04 mA cm^{-2}) than for the BE system (0.10 mA cm^{-2}), indicating significantly enhanced interfacial passivation (Fig. 5f). Post-cycling SEM images further confirmed the reversible Na deposition enabled by DDSLE electrolyte, with the Na foil remaining smooth after 500 cycles, in contrast to the mossy and degraded surface observed in BE electrolyte (Fig. 5g).⁴²

The synergistic optimization of both positive electrode and negative electrode interfaces by the DDSLE electrolyte significantly improves the cycling stability of Na-metal batteries. At 100 mA g^{-1} over 500 cycles, DDSLE-involved cells exhibited negligible capacity fade, contrasting sharply with BE-involved cells (48.5% retention; inset of Fig. 6a). The DC impedance of DDSLE-involved cell remained stable at $\sim 120 \Omega$, while BE-involved cell impedance rose continuously to 290Ω , indicating reduced electrolyte consumption and a stable interphase. Remarkably, at 1000 mA g^{-1} over 16,500 cycles, DDSLE-involved cell achieved 88.2% capacity retention with an average CE of 99.9%. This performance sets a benchmark for cycle life and degradation rate in high-voltage NFS positive electrodes operating to 4.5 V (Fig. 6b, Supplementary table 3). Even when compared to batteries using other positive electrode materials, such as polyanionic compounds (PACs), Prussian blue analogues (PBAs), and transition metal oxides (TMOs), which typically operate at lower voltages (3.8–4.3 V), the DDSLE-based Na||NFS cell still demonstrates fine cycling stability. Furthermore, DDSLE electrolyte enables rate capability, delivering a discharge specific capacity of 56.3 mAh g^{-1} at 5000 mA g^{-1} (Supplementary Fig. 19). While using NaDFOB as the sole electrolyte salt features a high anodic stability limit, this advantage is negated by the formation of an uneven and mechanically unstable interphases, which prevents long-term high-voltage operation and limits the cycle life to only about 1500 cycles (Supplementary Fig. 20, Supplementary Data 8 and note 5).

Post-mortem analysis confirmed the interfacial integrity preserved by the DDSLE electrolyte, with electrodes

and particles retaining their structure and separators remaining clean. In contrast, BE cells showed severe electrode pulverization, thick particle coatings, and brown deposits on separators (Fig. 6c), reflecting the baseline electrolyte's thermodynamic instability and the resulting cycle of sustained interfacial degradation it triggers. The minimal byproduct formation in DDSLE-involved cells signifies a stabilized internal environment during cycling, which is attributed to the DDSLE's ability to ensure reversible Na^+ transport by stabilizing the interface, suppressing electrolyte decomposition, and preventing byproduct accumulation. This performance stems from the elaborate solvation structure design, which redirects interfacial formation toward an orderly, controlled passivation guided by a specific anion, achieving synergistic reinforcement at both the thermodynamic (enhancing intrinsic stability) and kinetic (constructing a stable interphase) levels. Consequently, DDSLE effectively resolves issues of overcharge and rapid capacity fade at high voltages, with stability extending to 60 °C operation, where Na||NFS cells retain 95.2 % capacity after 180 cycles (Fig. 6d). Deposits on BE-involved separators reveal significant Fe dissolution (Supplementary Fig. 21), indicating byproduct-catalyzed decomposition of polyanionic frameworks within separator pores. Crucially, the boride/fluoride-rich CEI in DDSLE electrolyte further inhibits Fe dissolution/migration and enhances interfacial stability.

As-assembled Na||NFS pouch cells with DDSLE electrolyte exhibits a discharge capacity of 1.57 mAh cm^{-2} at 100 mA g^{-1} , retaining 93.9% capacity after 500 cycles with a stable average discharge voltage of 3.72 V (Fig. 6e, Supplementary Fig. 22). Based on total electrode mass, the specific energy reached 156.1 Wh kg^{-1} . Compared to literature reports for Na-metal pouch cells, DDSLE electrolyte achieves competitive comprehensive performance, including higher operating voltage, enhanced cycle stability, improved safety, compatibility with polypropylene (PP) separators, and minimized interfacial side reactions (Fig. 6f, Supplementary table 4).^{43–45}

DDSLE electrolyte demonstrates good compatibility with diverse positive electrodes across voltage windows of 2.0–4.0 V, 2.0–4.2 V, and 2.0–4.5 V (Fig. 6g). While $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ (NVP) and $\text{Na}_{0.72}\text{Ni}_{0.32}\text{Mn}_{0.68}\text{O}_2$ (P2-TMO) positive electrodes show comparable discharge capacities in both electrolytes, DDSLE electrolyte substantially reduces irreversible capacity loss, increasing Coulombic efficiencies from 93.1% to 99.6% at 4.0 V and 69.4% to 97.1% at 4.2 V. For the high-voltage NFS positive electrode, CE improves from 89.0% to 99.6% at 4.5 V. After 50 cycles, capacity retention increases by 11.1% (Na||NVP), 25.2% (Na||P2-TMO), and 2.3% (Na||NFS), with corresponding average CE enhancements of 19.3%, 24.4%, and 24.3% (Fig. 6h). The DDSLE-supported stability for diverse positive electrodes over longer cycles is also demonstrated in Supplementary Fig. 23. Furthermore, the dual-domain molecular-locked strategy demonstrates broad applicability across various battery systems, achieving markedly elevated anodic stability limits in diverse electrolyte formulations (Supplementary Fig. 24). These results confirm DDSLE's efficacy in forming electrochemically stable interphases and extending battery cycling lifetimes.

In summary, we provide a dual-domain solvent-locked electrolyte (DDSLE) to resolve the persistent instability challenge of industrial NaPF_6 /carbonate-based electrolytes at high voltage (>4.3 V). By integrating a symmetrical difluoro(oxalato)borate anion (DFOB^-), a solvent-reinforced Na^+ solvation within bulk solution (domain I) and an anion-rich shield at positive electrode interface (domain II) are constructed, effectively restricting the freedom degrees and reactivity of solvent molecules. We called it, therefore, as dual-domain solvent-locked electrolyte. This solvent-locked configuration shifts electrolyte decomposition pathways from solvent- to anion-dominance, fostering stable interphases. On positive electrode side, an integrated CEI with a boride/fluoride-rich inner and polyester/alkoxide-rich outer distribution effectively suppresses electron tunneling, reducing positive electrode contact potential by 16.7% and leakage current by 34.6–79.2%. Combined experimental and theoretical analyses establish and validate a linear correlation between decomposition voltage and the bandgap of CEI components, identifying electron tunneling

resistance as the primary determinant of high-voltage interfacial stability. Critically, borides/fluorides with bandgaps exceeding 6.2 eV are demonstrated to effectively suppress electron tunneling above 4.0 V. Concurrently, a thin, uniform SEI forms on Na negative electrode, enabling reversible Na deposition. These synergistic effects significantly enhanced electrolyte stability at high voltages, evidenced by an increased anodic stability limit (from 3.4 V to 5.1 V), improved Coulombic efficiency, reduced over-charging and gaseous release (46.6% reduction). The DDSLE electrolyte demonstrates compatibility with a sodium negative electrode and diverse positive electrodes, including $\text{Na}_{2.26}\text{Fe}_{1.87}(\text{SO}_4)_3$, $\text{Na}_3\text{V}_2(\text{PO}_4)_3$ and $\text{Na}_{0.72}\text{Ni}_{0.32}\text{Mn}_{0.68}\text{O}_2$. The assembled $\text{Na}||\text{Na}_{2.26}\text{Fe}_{1.87}(\text{SO}_4)_3$ cell delivers long-term cycling stability at a cut-off voltage of 4.5 V, achieving 88.2% capacity retention after 16,500 cycles in coin cell and 93.9% retention after 500 cycles in pouch cell. Our work opens a promising avenue for systematic electrolyte engineering in next-generation high-voltage batteries.

Methods

Materials

Battery-grade ($\geq 99.8\%$) NaPF_6 , EC, EMC, and NaDFOB were received from Duoduo chemical reagent Co., Ltd. P2-TMO was received from Paragonage Technology Co., Ltd. NVP, conductive carbon (super P Li, MA-EN-CO, $>99.5\%$ purity; Ketjen black, ECP-600JD, $>99.8\%$ purity), polyvinylidene fluoride (PVDF, ARKEMA, molecular weight of $\sim 1,000,000$), PP (Celgard 2500, thickness: 25 μm , pore size: 0.064 μm , porosity: 55%) and glass fiber (Whatman GF/D, thickness: 260 μm , pore size: 2.7 μm , porosity: 40–50%) membranes, N-methyl-2-pyrrolidone (NMP, $>99.9\%$ purity), and Al foil (Al $>99.999\%$, 20 μm) were bought from Canrd New Energy Technology Co., Ltd. Anhydrous sodium sulfate (Na_2SO_4 , $\geq 99.0\%$), iron sulfate (FeSO_4 , $\geq 99.0\%$), ascorbic acid ($\text{C}_6\text{H}_6\text{O}_6$, $\geq 99.7\%$), and citric acid ($\text{C}_6\text{H}_{10}\text{O}_8$, $\geq 99.5\%$) were bought from Sinopharm Chemical Reagent Co., Ltd. without further purification. Graphene oxide powder was bought from Shenghui's New carbon material store on the Taobao website.

Preparation of electrolyte and positive electrode materials

Prior to use, pretreat the solvents by immersing activated molecular sieves (calcined at 500 $^\circ\text{C}$ for 8 hours) in them for 48 hours to remove residual water. Prepare a mixed solvent of EC and EMC in a volume ratio of 1:1. Then, dissolve one or more sodium salts into the solvent mixture at specified molar ratios to formulate high-throughput batch electrolytes. Specifically, the baseline electrolyte (BE) is prepared by dissolving 1M NaPF_6 in EC/EMC. The dual-domain solvent-locked electrolyte (DDSLE) is formulated by further adding 0.4M NaDFOB to the BE electrolyte.

The NFS was synthesized via a modified co-precipitation route initiated by dispersing 0.3 g graphene oxide and 0.3 g Ketjen black in 60 mL deionized water under ultrasonication (100 W, 30 min; KQ-250DS, Kunshan Ultrasonic Instruments Co. Ltd.). To this dispersion, stoichiometric precursors (12 mmol Na_2SO_4 , 18 mmol FeSO_4), along with 0.3 g citric acid and 0.12 g ascorbic acid were added under continuous stirring. The mixture underwent low-temperature co-precipitation ($<8^\circ\text{C}$) via dropwise addition of 120 mL isopropanol, followed by sand milling (2000 rpm, 1 hour; nano sanding machine 30 L, Kunshan Zhuowei Machinery Technology Co. Ltd.) to homogenize particle size. Subsequent spray-drying at 300 mL/hour (TENLIN-6000Y, Jiangsu Tianling Instrument Co. Ltd.) yielded a precursor that was calcined in Ar at 350 $^\circ\text{C}$ for 12 hours (2 $^\circ\text{C}/\text{min}$ ramp) to crystallize the final NFS material.

Cell assembly and electrochemical measurements

The positive electrode slurry was prepared by mixing the active materials (NFS, NVP, NFPP, P2-TMO), carbon (super P Li), and PVDF binder in a mass ratio of 8:1:1, using NMP as the dispersing solvent. The slurry was then coated onto an Al foil using a doctor blade. The coated electrode was dried at 120 $^\circ\text{C}$ under vacuum for 10 hours,

followed by punching into circular electrodes with a diameter of 12 mm. The average mass loading and thickness of the electrodes were approximately 1.8–2.6 mg cm⁻² and 33–47 μm, respectively.

All cell assembly procedures were conducted in an argon-filled glove box, with both H₂O and O₂ levels maintained below 1 ppm. For the assembly of Na-metal coin cells, the configuration was as follows: sodium metal (70-μm thick) as the negative electrode (16 mm in diameter), and a glass fiber membrane (19 mm in diameter) as the separator. The added amount of electrolyte was 120 μL per cell. The assembled cell was then sealed using a hydraulic coin cell crimping machine to obtain a CR2025-type coin cell. For the assembly of Na-metal pouch cells, the configuration was as follows: sodium metal (70-μm thick) as the negative electrode (56*43 mm), NFS as the positive electrode (7~9 mg cm⁻², 56*43 mm, 89-114 μm.), and a PP membrane as the separator. The added amount of electrolyte was 150 μL per positive electrode sheet. Before the electrochemical cycle begins, the pouch cell is cycled at 20 mA g⁻¹ for 3 cycles. The cycled samples (washed with EMC solvent and dry in argon-filled glovebox) were sealed in an Ar-filled vessel and transferred into the chamber through a sealed plastic tube.

Galvanostatic charge and discharge measurements were carried out at Neware CT-4008 battery testing system (Shenzhen, China). EIS and CV tests were performed on an electrochemical workstation (Autolab RRDE/RDE-2) in an oven (DH2500B incubator, Tianjin Tongli Xinda Instrument Factory). EIS measurements were performed with the potentiostatic signal, and the amplitude of the AC voltage was set at 5 mV at 100 kHz to 0.01 Hz. Unless otherwise specified, all cells function in a constant-temperature environment (25 ± 1 °C).

***In situ* characterizations**

In situ mass spectrometry. Positive electrodes (active material loading: 1.0–1.5 mg cm⁻²) were assembled in an Ar-filled glovebox using a dedicated *in situ* mold (Beijing Scistar Technology Co., Ltd.) with Na metal counter electrodes and 16 mm glass fiber separators. Following the addition of 200 μL electrolyte, cells were charged and discharged at 50 μA cm⁻². Pre-dehydrated He (99.999%) served as the carrier gas; the gas line was purged at high flow to remove residual air before stabilizing at 1 mL/min via a mass flow controller. Evolved gases passed through a low-temperature condenser to trap electrolyte volatiles prior to detection by an HPR-20 DEMS system (Hidden Analytical).

In situ electrochemical quartz crystal microbalance (*in situ* EQCM). NFS powder was dispersed in PVDF/NMP solution (3 wt.%, 5 mL) to form a suspension. Following 30 min sonication, 20 μL supernatant was drop-cast onto a silicon wafer and dried under vacuum (120 °C, 4 h). *In situ* cells were assembled in an Ar-filled glovebox using a specialized mold with Na metal negative electrodes and 200 μL electrolyte. EQCM measurements were performed on a QSense Explorer (Biolin Scientific AB) using Au-coated quartz crystal sensors (QSX 313). The Au-coated sensor served as the working electrode; charge–discharge cycling employed a constant potential increment mode (50 mV s⁻¹, 2.0–4.5 V vs. Na⁺/Na, 100 cycles).

In situ Raman. Positive electrodes (active material loading: 1.0–1.5 mg cm⁻²) were assembled with Na metal counter electrodes and glass fiber separators in custom cells (Beijing Scistar Technology Co., Ltd.) containing 200 μL electrolyte. Spectra were collected using a HORIBA Scientific Lab-RAM HR Evolution system (532 nm excitation laser) with a 10 min scan interval. Charge–discharge processes were controlled by a Lvium electrochemical workstation at 50 μA cm⁻².

All *in situ* characterizations are conducted in a constant temperature environment (25 ± 1 °C).

Materials characterizations

Morphologies were characterized by field-emission SEM (ZEISS Merlin Compact) equipped with EDS (Oxford X-Max) and TEM (FEI Tecnai G2 F20 S-TWIN). Frequency-modulated high-resolution AFM was performed using a Shimadzu SPM-8100FM. FT-IR spectra were recorded on a Bruker INVENIO S spectrometer. X-ray photoelectron

spectrometer (XPS, Thermo ESCALAB MK II) with an X-ray source (Al K α , $h\nu = 1486.6$ eV, all the peaks were calibrated with the C 1s spectrum at a binding energy of 284.8 eV). Backgrounds were subtracted using the Shirley method, and peaks were fitted with mixed Gaussian–Lorentzian functions. Depth profiling utilized Ar⁺ sputtering (2 kV, 2 μ A, 45° incident angle). TOF-SIMS (ION-TOF GmbH) was performed on cycled electrodes (EMC solvent-washed, Ar-dried) using a pulsed Bi³⁺ ion beam (30 eV, 20 ns) for depth profiling and a Cs⁺ ion beam (500 eV) for surface analysis. NMR spectra were obtained on a Bruker Advance spectrometer. Gas evolution was monitored via optical microscopy (LIB-MS750). Synchrotron high-pressure powder XRD was conducted at ANSTO, Australia ($\lambda = 0.6887$ Å); Rietveld refinement was performed using GSAS-II software.

Cryo-TEM was performed using an FEI Glacios (USA) operated at 200 kV, which is equipped with an automatic injection system for frozen samples. HRTEM was conducted using a JEOL JEM-2100 operated at 200 kV. For *ex situ* measurements, the electrodes were washed three times with EMC solvent and then dried for 12 h at 25 °C in an argon-filled glovebox (O₂ and H₂O < 0.1 ppm). Electrochemical cycling was carried out in Na||electrode coin cells at 10 mA g⁻¹. After 3 cycles, the cells were disassembled at a discharged state of 2.0 V. The harvested electrodes were rinsed with EMC to remove residual electrolyte salts, followed by vacuum drying in the argon-filled glovebox. All sample preparation steps were conducted under an argon atmosphere. For cryo-TEM, the grids were placed in a centrifuge tube and transferred to the TEM column under liquid nitrogen. For HRTEM, the grids were quickly transferred to the sample stage and the column was evacuated immediately. Electrode samples were transported from the argon-filled glovebox to the testing facility using vacuum-sealed plastic device to prevent air exposure.

Computational details

DFT calculations for electrolyte components were performed using Gaussian 16 (Frisch, M.J. et al. Gaussian, Inc., Wallingford CT, 2016). Geometries of molecules and solvation shells were optimized at the B3LYP/6-31+G(d,p) level, with frequency analyses confirming all structures as stationary points. Periodic DFT calculations for electrolyte/Na composite models were conducted using the Vienna Ab initio Simulation Package (VASP).⁴⁶ Exchange-correlation interactions were described using the Perdew–Burke–Ernzerhof (PBE) functional within the generalized gradient approximation (GGA) and the projector augmented wave (PAW) method. A plane-wave cutoff energy of 400 eV was employed, with energy and force convergence criteria set to 10⁻⁵ eV and 0.02 eV/Å, respectively. Brillouin zone sampling utilized G-centered k-point meshes.

MD simulations were carried out using Gromacs 2020 with the all-atom optimized potentials for liquid simulations (OPLS-AA) force field.⁴⁷ The force field parameters were generated using LigParGen web server. The model cells of two electrolyte contain 20 Na⁺ + 150 EC + 97 EMC + 20 PF₆⁻ for (BE electrolyte), and 25 Na⁺ + 150 EC + 97 EMC + 20 PF₆⁻ + 5 DFOB⁻. Initial structures were relaxed by energy minimization, NVT, NPT, and MD process. Velocity-rescale thermostat with a relaxation constant of 1 ps was used to control the temperature at 298.15 K. An MD simulation for a total simulation time of 2 ns was then performed, and the trajectory was saved every 1 ps. For the positive electrode-electrolyte composite model construction and kinetic optimization, MD simulations were performed using the CP2K package (version 7.0, <https://www.cp2k.org/>). The simulation system contained a conventional unit cell of Na_{2.26}Fe_{1.87}(SO₄)₃ exposing the (200) crystal plane. The simulations employed Gamma-centered k-point sampling with a plane-wave energy cutoff of 450 eV, and the total energy was converged to 10⁻⁵ eV. The DFT-relaxed structure was thermally equilibrated in the NVT ensemble using a 1-fs timestep and a Nosé-Hoover thermostat (Nosé-mass parameter: 500 timesteps), with stepwise heating from 100 K to the target temperature of 300 K.

Data availability

All data supporting this study and its findings are available within the article and Supplementary Information. Additional information can be obtained from the corresponding authors upon request. Source data are provided with this paper.

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Author Contributions

W.C. conceived the idea and supervised the project. J.Z. and G.T. designed and conducted the experiments. S.M. conducted *in situ* electrochemical characterizations. Y.C. participated in electrolyte preparation and conducted electrochemical performance. L.W. and J.L. performed the theoretical calculations. X.G., C.H. and E.W. participated in discussion. J.Z., G.T. and W.C. wrote and revised the manuscript. All authors discussed the results and commented on the manuscript. J.Z. and G.T. contributed equally to this work.

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Competing interests

The authors declare no competing interests.

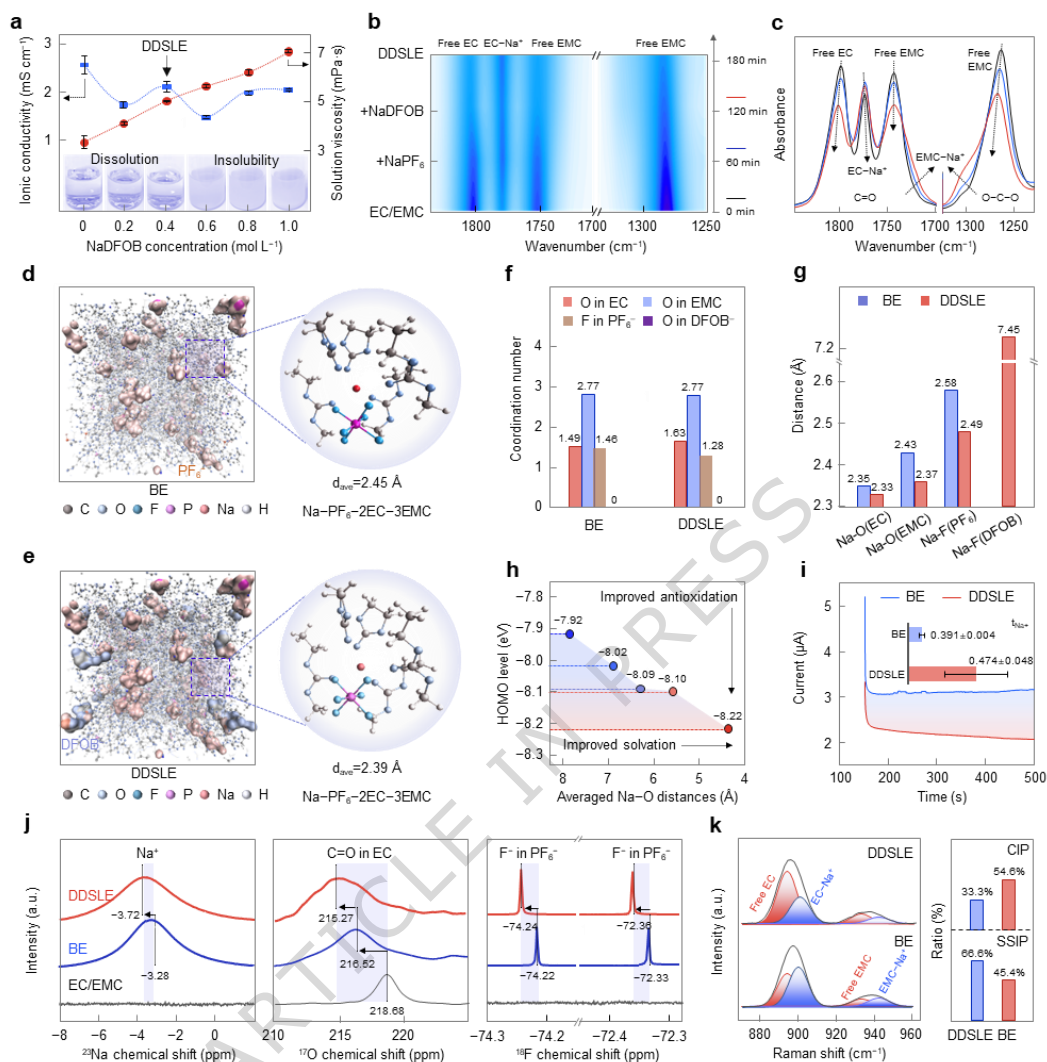


Figure 1 | Chemistry and solvation structure of DDSLE electrolyte. (a) Ionic conductivities, solution viscosity, digital images of electrolytes with varying NaDFOB concentrations. (b, c) *In situ* FTIR spectra tracking salt dissolution in EC/EMC from the BE to the DDSLE electrolytes. Snapshots from MD simulations for the (d) BE and (e) DDSLE electrolytes, depicting PF₆⁻ (yellow isosurface) and DFOB⁻ (blue isosurface); Na⁺ ions and EC/EMC molecules are shown in stick representation. (f) Average coordination numbers and (g) Na-O/F bond distances obtained from MD simulations. (h) Evolution of the HOMO energy for the Na⁺ solvation structure as a function of coordination distance. (i) Chronoamperometry profiles of BE and DDSLE electrolytes in Na||Na symmetric cells. Inset: calculated Na⁺ transference number with error bar. (j) ²³Na, ¹⁷O, and ¹⁹F NMR spectra of the pure solvents, BE, and DDSLE electrolytes. (k) Raman spectra of the BE and DDSLE electrolytes, alongside the corresponding quantification of solvation species. The error bar represents the standard deviation of the fitted curve, with data derived from three parallel experiments.

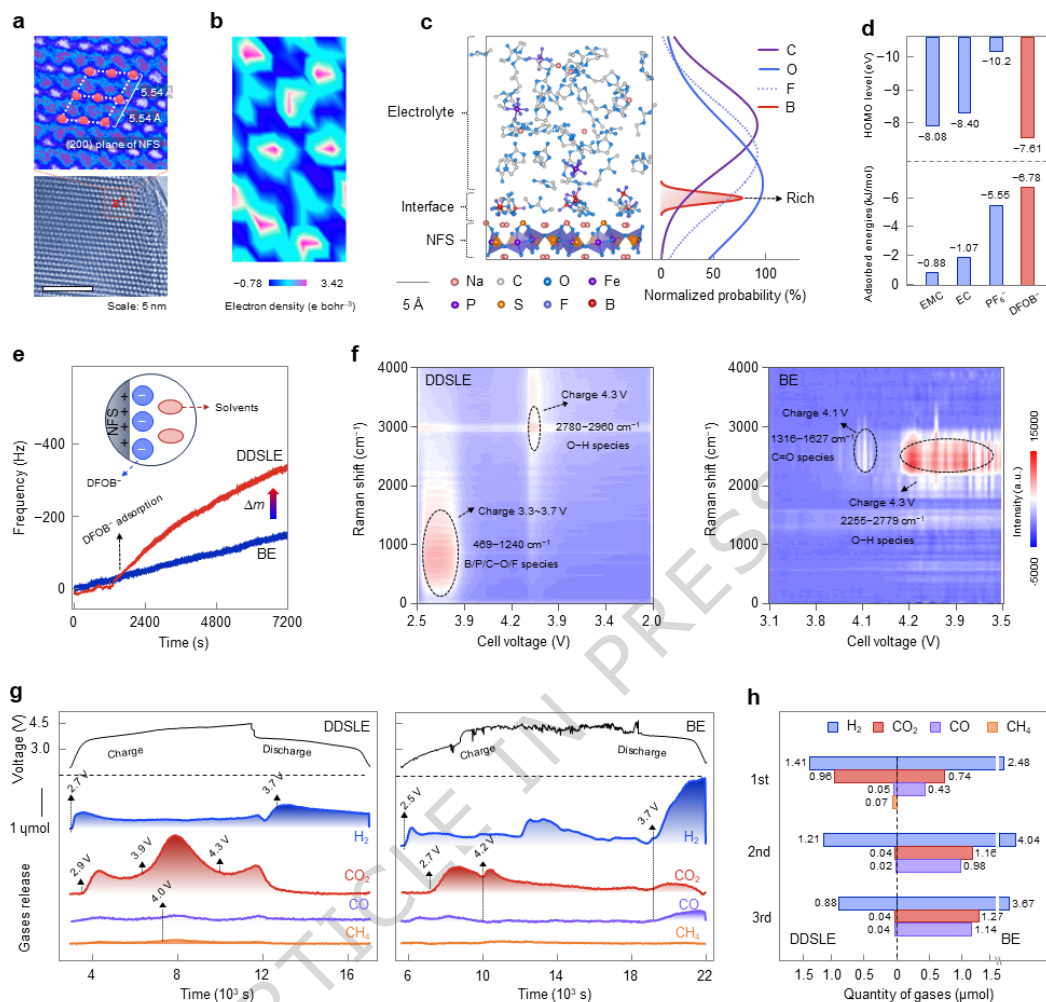


Figure 2 | Structural evolution and reactions of DDSLE electrolyte at positive electrode interface. (a) HRTEM images of NFS positive electrode. (b) Electron density map of NFS (200) plane. (c) NFS/DDSLE interfacial model, with distribution profiles of anions and solvent molecules along *c*-axis of the simulation box. (d) Calculated adsorption energies of electrolyte components on NFS surface and HOMO energy levels. (e) Time-dependent oscillation frequency shifts of a Na||NFS cell at 50 mV s⁻¹ and 25 °C, measured by EQCM. (f) *In situ* Raman spectra of Na||NFS cells employing DDSLE and BE electrolytes at 50 μA cm⁻² and 25 °C. (g) *In situ* DEMS data for Na||NFS cells with DDSLE and BE electrolytes during the initial charging process to 4.5 V at 50 μA cm⁻² and 25 °C. (h) Calculated gaseous release amounts.

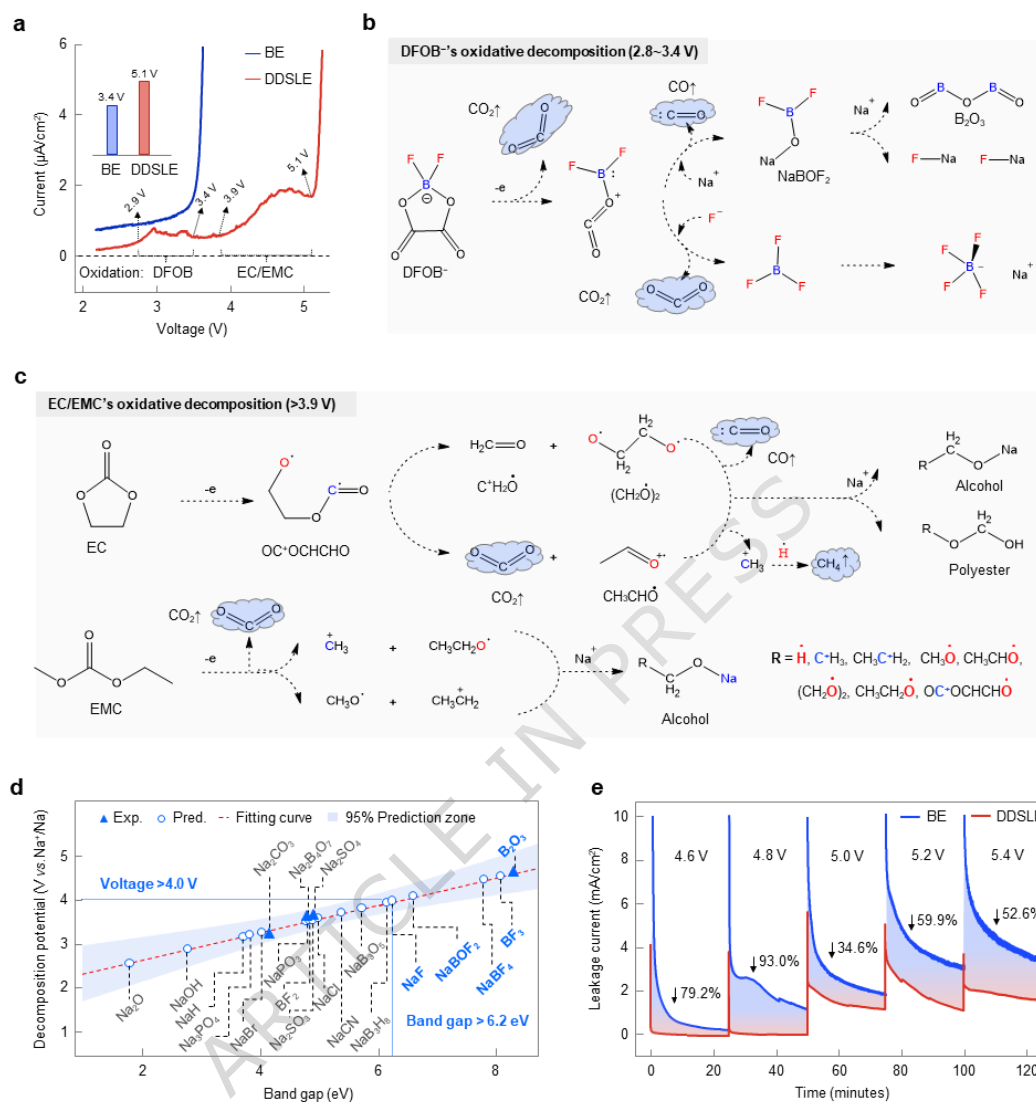


Figure 3 | Suppressed electrochemical reactivity of DDSLE electrolyte on NFS positive electrode. (a) LSV curves of Na||stainless steel sheet cells. Proposed chemical transformations during the oxidative decomposition of the (b) DFOB⁻ anion (between 2.8 and 3.4 V) and (c) EC/EMC solvent molecules (above 3.9 V). (d) Decomposition potentials of typical inorganic components in CEI and their corresponding band gap values. (e) Leakage current of Na||NFS cells with DDSLE and BE electrolytes during charging to high voltages (>4.5 V).

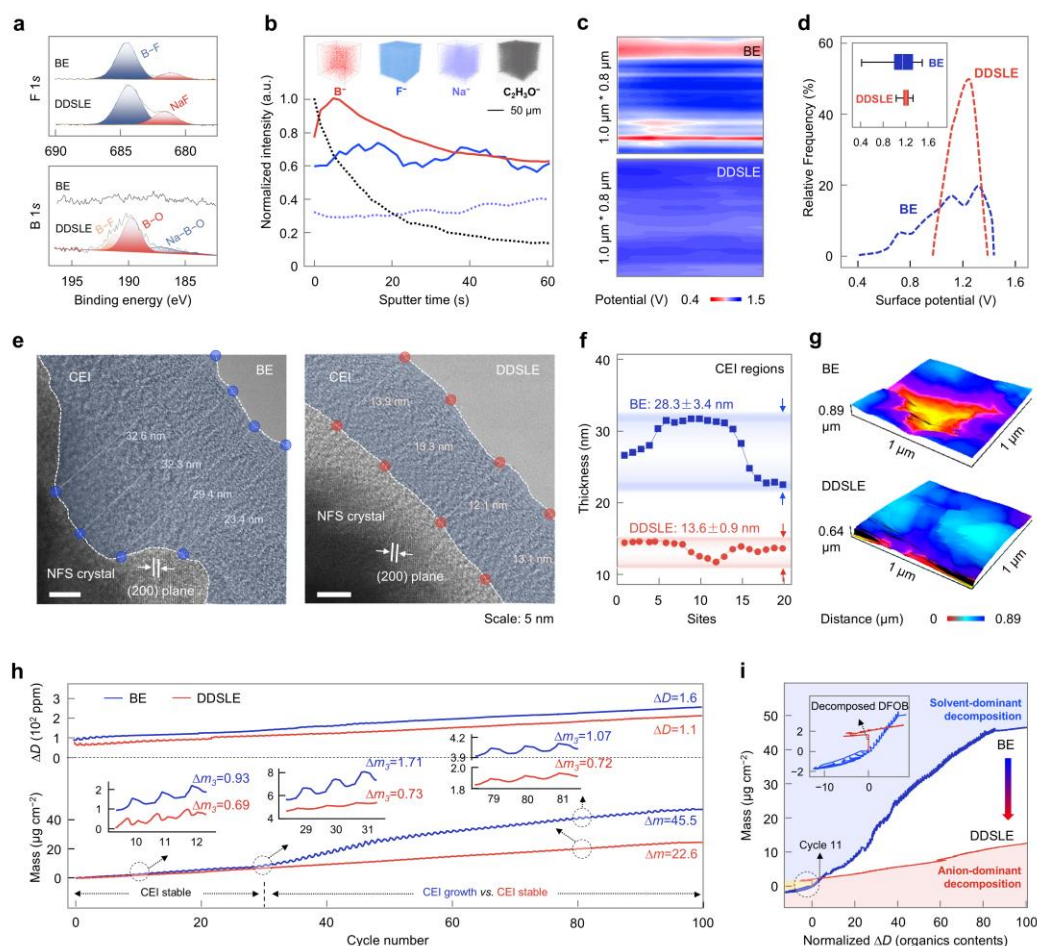


Figure 4 | Composition, structure and mechanical properties of CEI. (a) B 1s and F 1s XPS spectra of cycled NFS positive electrodes. (b) TOF-SIMS 3D reconstruction and relative intensities of B⁺, F⁻, Na⁺, and C₂H₃O⁻ species as a function of sputtering depth for the DDSLE-derived CEI. (c, d) Electrostatic potential mapping of cycled NFS positive electrodes using DDSLE and BE electrolytes. (e) Cryo-TEM images of the CEI formed with BE and DDSLE electrolytes. (f) Thickness distribution of the CEI. (g) Surface roughness mapping of cycled NFS positive electrodes with DDSLE and BE electrolytes. (h) Time-dependent changes in mass (Δm) and energy dissipation (ΔD) for a Na||NFS cell over multiple cycles during CV measurement at 50 mV s⁻¹ and 25 °C recorded via *in situ* EQCM. (i) Correlation between the mass change and energy dissipation. The electrodes in Figures a–g were tested in Na||NFS coin cells at 10 mA g⁻¹. After 3 cycles, the cells were disassembled at a discharged state of 2.0 V.

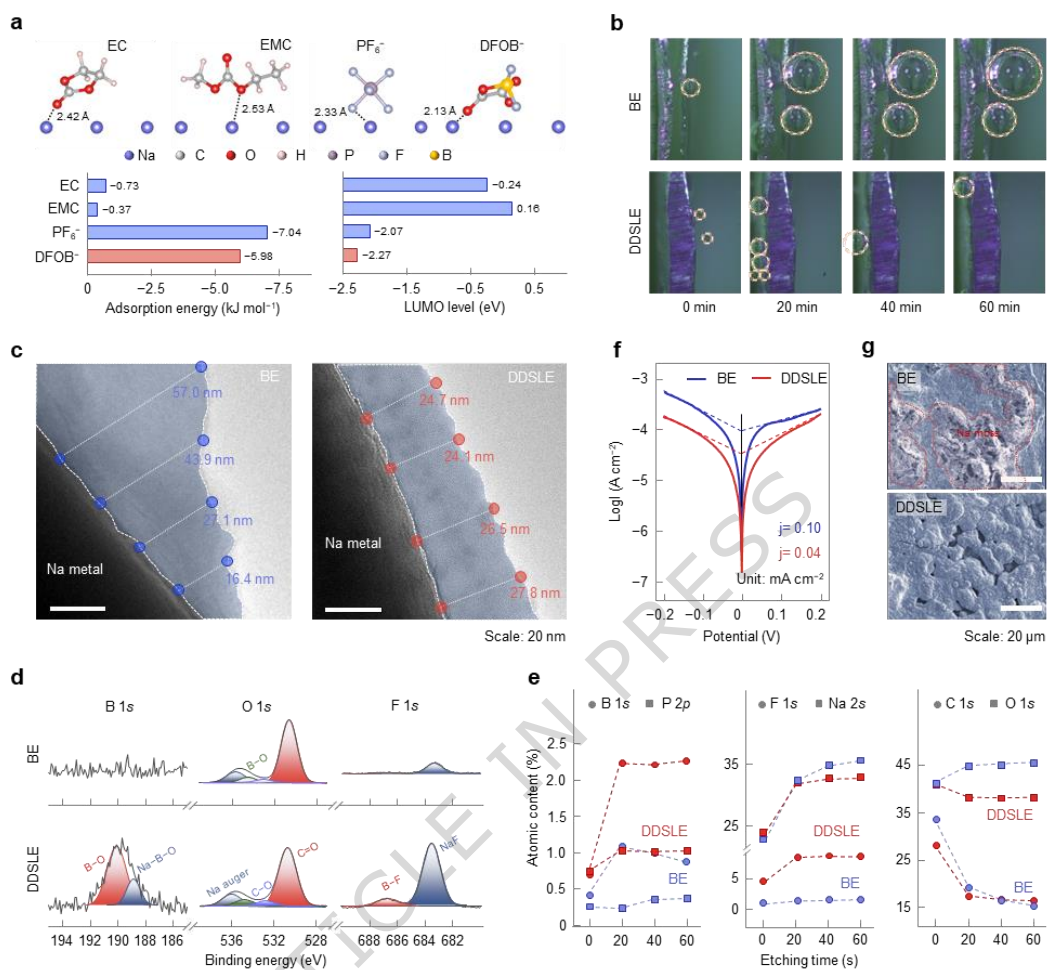


Figure 5 | Interfacial chemistry and SEI structure. (a) The structure of the electrolyte components adsorbed on Na metal surface, as well as their adsorption energy and LUMO level. (b) *In situ* optical microscopy images of the Na metal surface in Na||Na symmetric cells with DDSLE and BE electrolytes at 50 $\mu\text{A cm}^{-2}$ and 25 $^{\circ}\text{C}$. (c) TEM images of the SEI formed in BE and DDSLE electrolytes. (d) B 1s and F 1s XPS spectra of cycled Na negative electrodes. (e) Evolution of elemental composition as a function of XPS sputtering depth. (f) Tafel plots of Na||Na symmetric cells with DDSLE and BE electrolytes. (g) SEM images of the Na metal surface after 500 cycles in Na||NFS cells. The electrodes in Figures c, d, e and g were tested in Na||NFS coin cells at 10 mA g⁻¹. After 3 cycles, the cells were disassembled at a discharged state of 2.0 V.

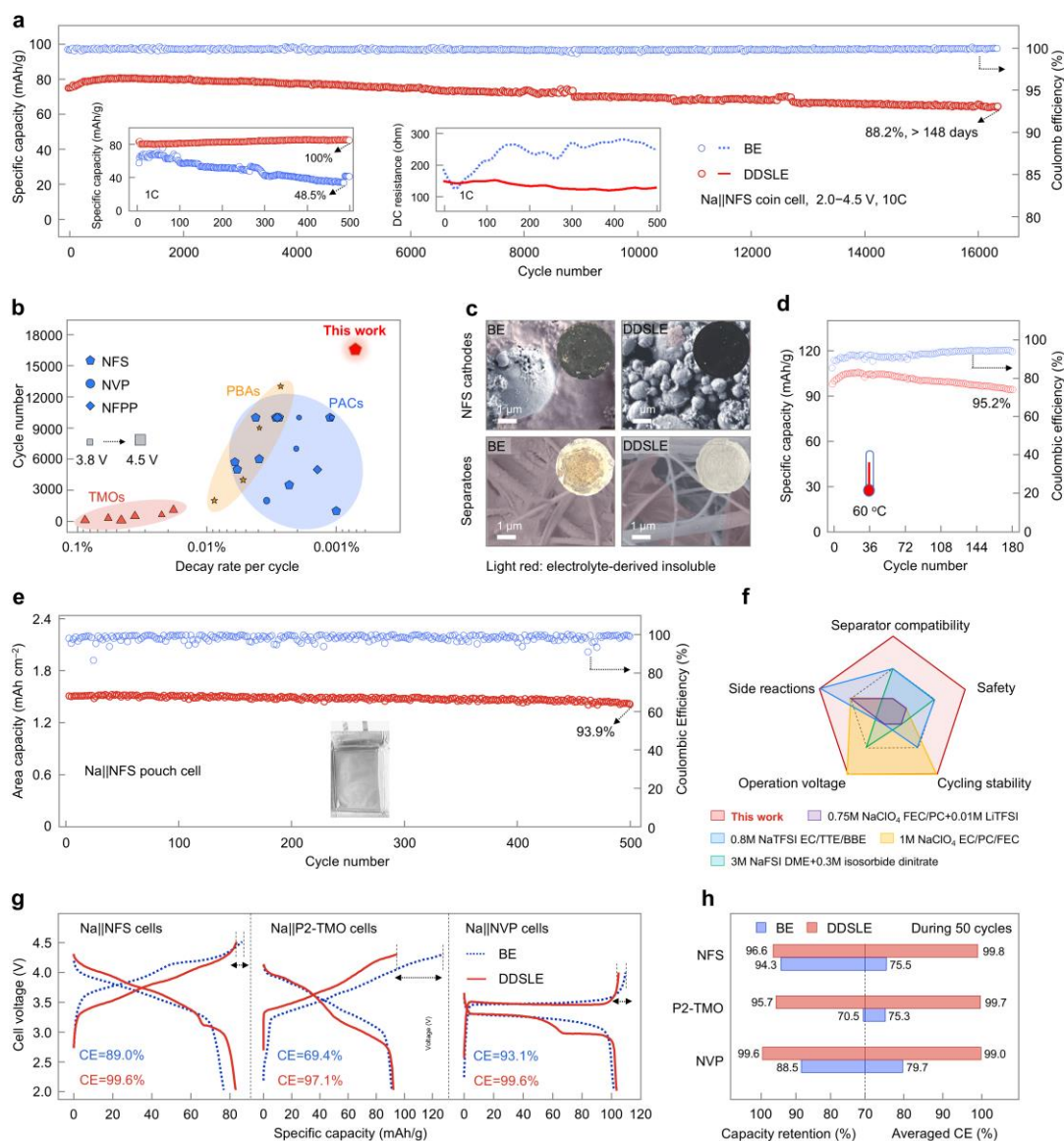


Figure 6 | Electrochemical performance of Na-metal coin and pouch cells with DDSLE electrolyte. (a) Long-term cycling performance of Na||NFS coin cells at 1000 mA g⁻¹ and 25 °C. Inset: cycling performance and DC internal resistance at 1000 mA g⁻¹ at 25 °C. (b) Benchmarking of Na||NFS cell against state-of-the-art Na-metal cells. The source of the literature data shown in this figure can be found in Supplementary Information, Table 3. (c) SEM images of the positive electrodes and separators from Na||NFS coin cells after 500 cycles at 1000 mA g⁻¹ and 25 °C. Inset: The image of the corresponding battery after disassembling the pole piece and separator. The electrode was tested in Na||NFS button cell at 1000 mA g⁻¹, and the cell was cycled for 500 cycles before disassembly and stopped at 2.0V. (d) Cycling performance of Na||NFS coin cells at 100 mA g⁻¹ and 60 °C. (e) Cycling performance of Na||NFS pouch cells at 100 mA g⁻¹ and 25 °C. (f) Radar charts comparing the performance of representative electrolyte in Na-ion cell systems. The source of the literature data shown in this figure can be found in Supplementary table 4. (g) Galvanostatic charge-discharge profiles of Na||NVP, Na||P2-TMO, and Na||NFS coin cells using DDSLE and BE electrolytes at 100 mA g⁻¹ and 25 °C. (h) Statistics of capacity retention and averaged Coulombic efficiency of cells during initial 50 cycles.

Editor Summary

Traditional electrolytes are electrochemically unstable on the positive electrode surface. Here, the author designed a solvent locked electrolyte to form a stable boride/fluoride interface. The assembled Na||Na_{2.26}Fe_{1.87}(SO₄)₃ battery maintained 88.2% capacity after 16500 cycles at 1 A g⁻¹.

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